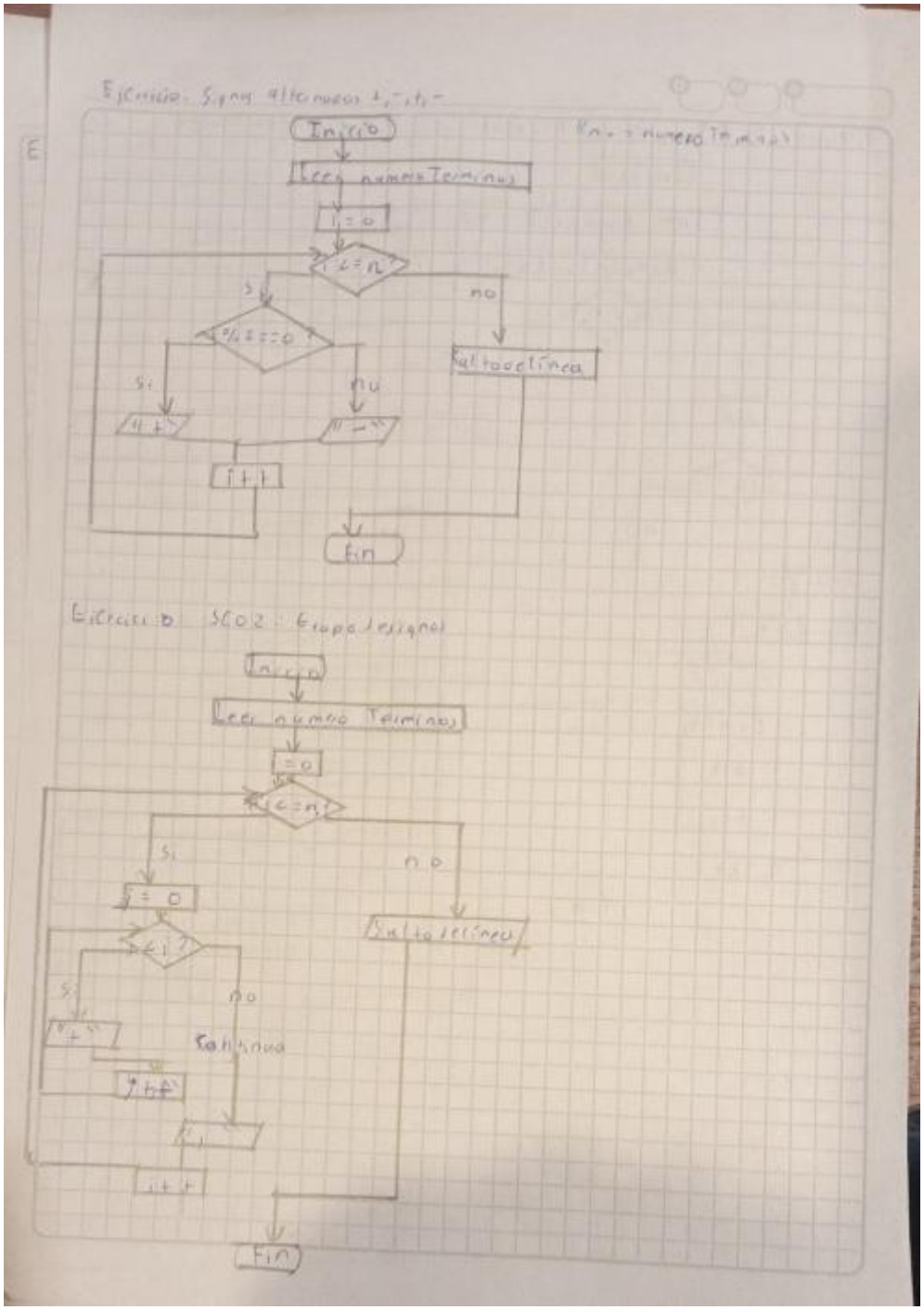
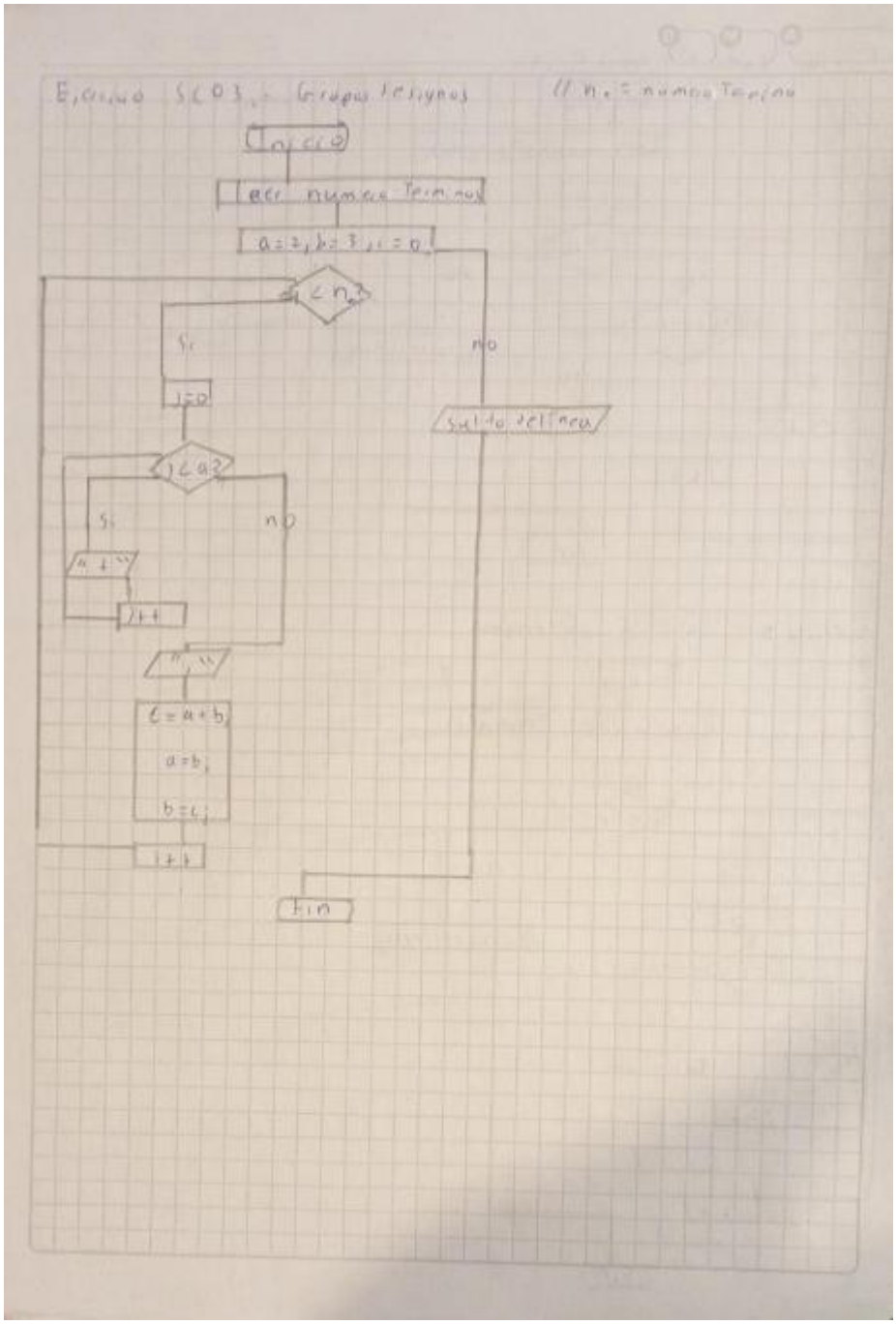
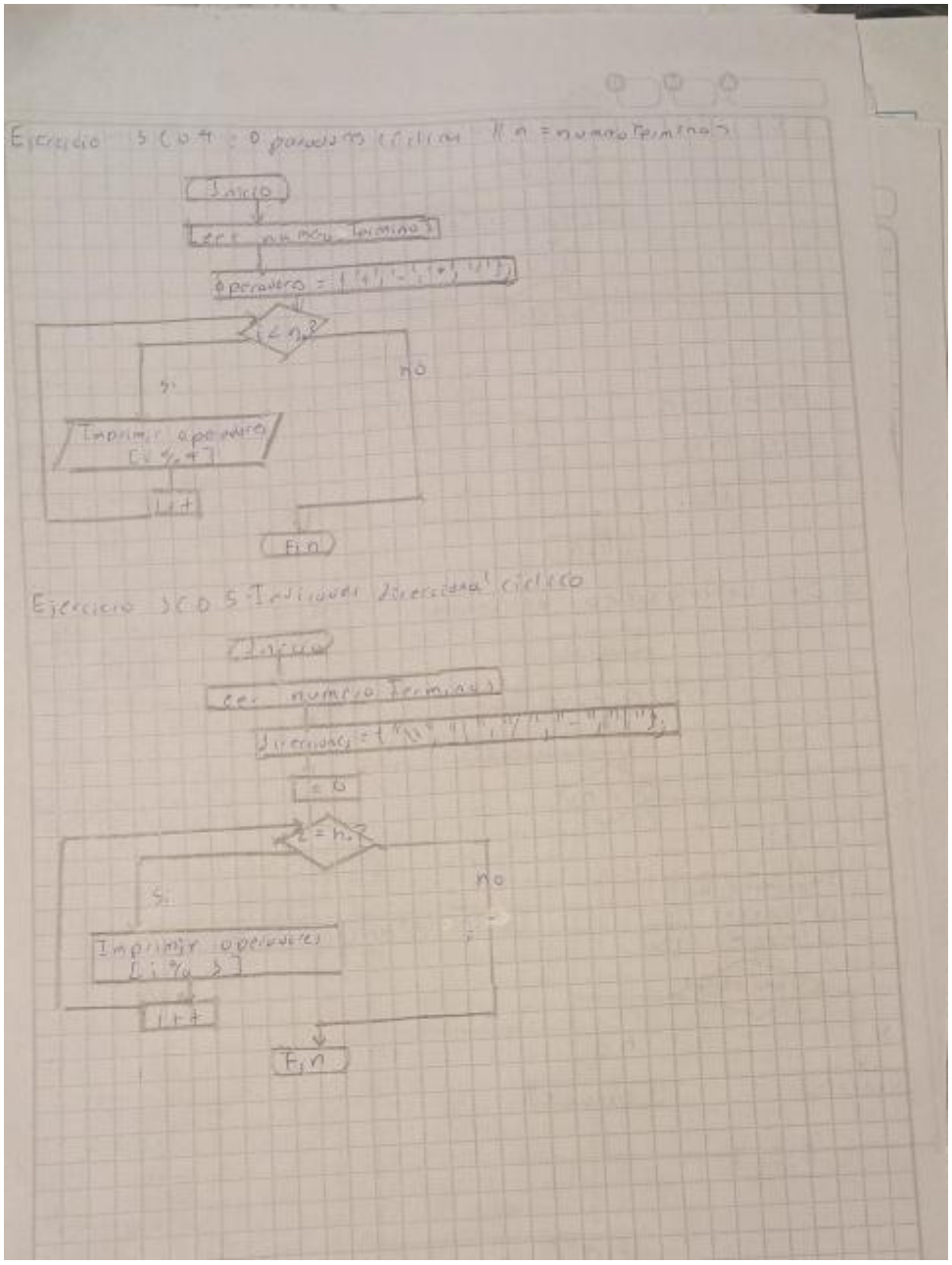
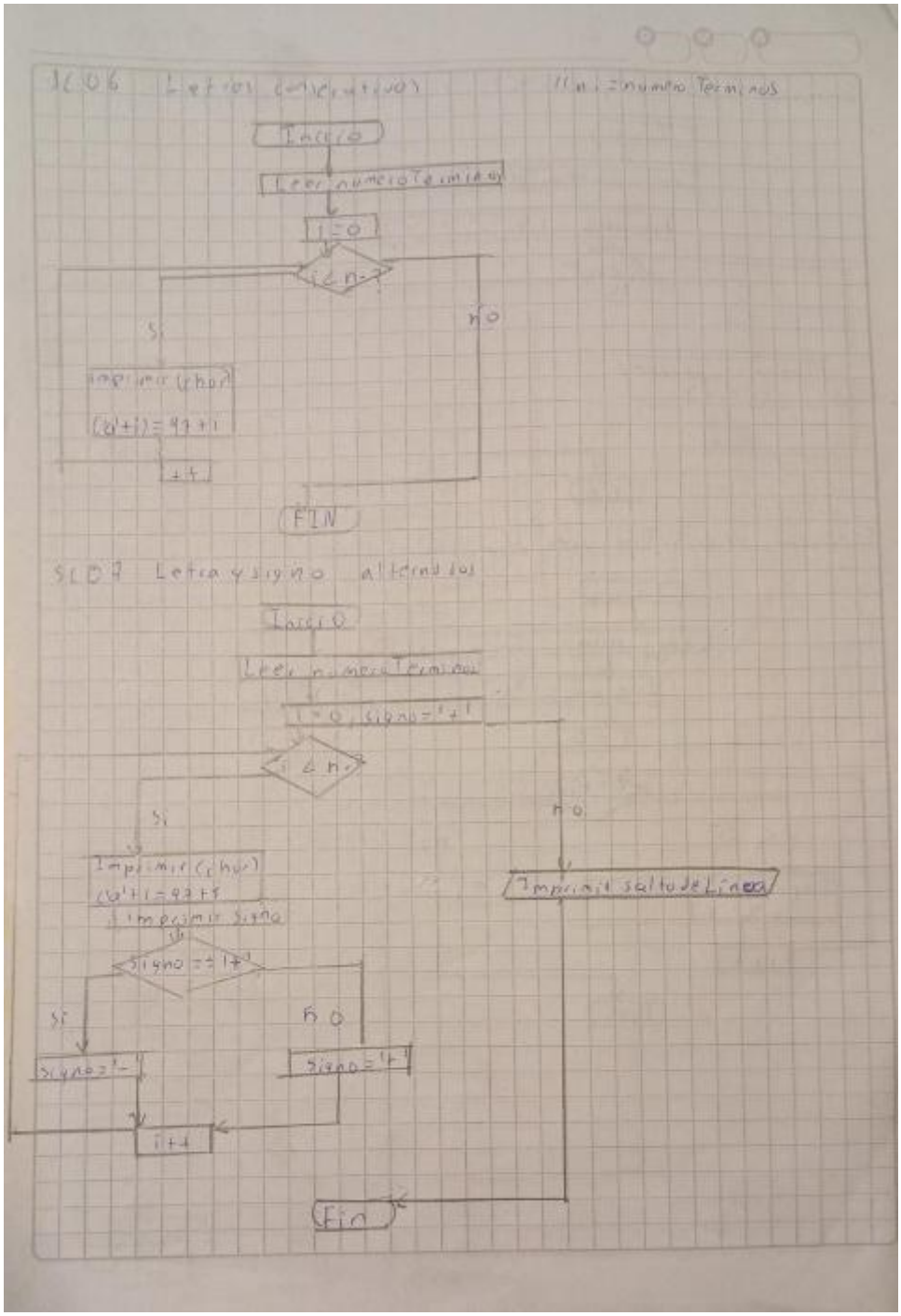
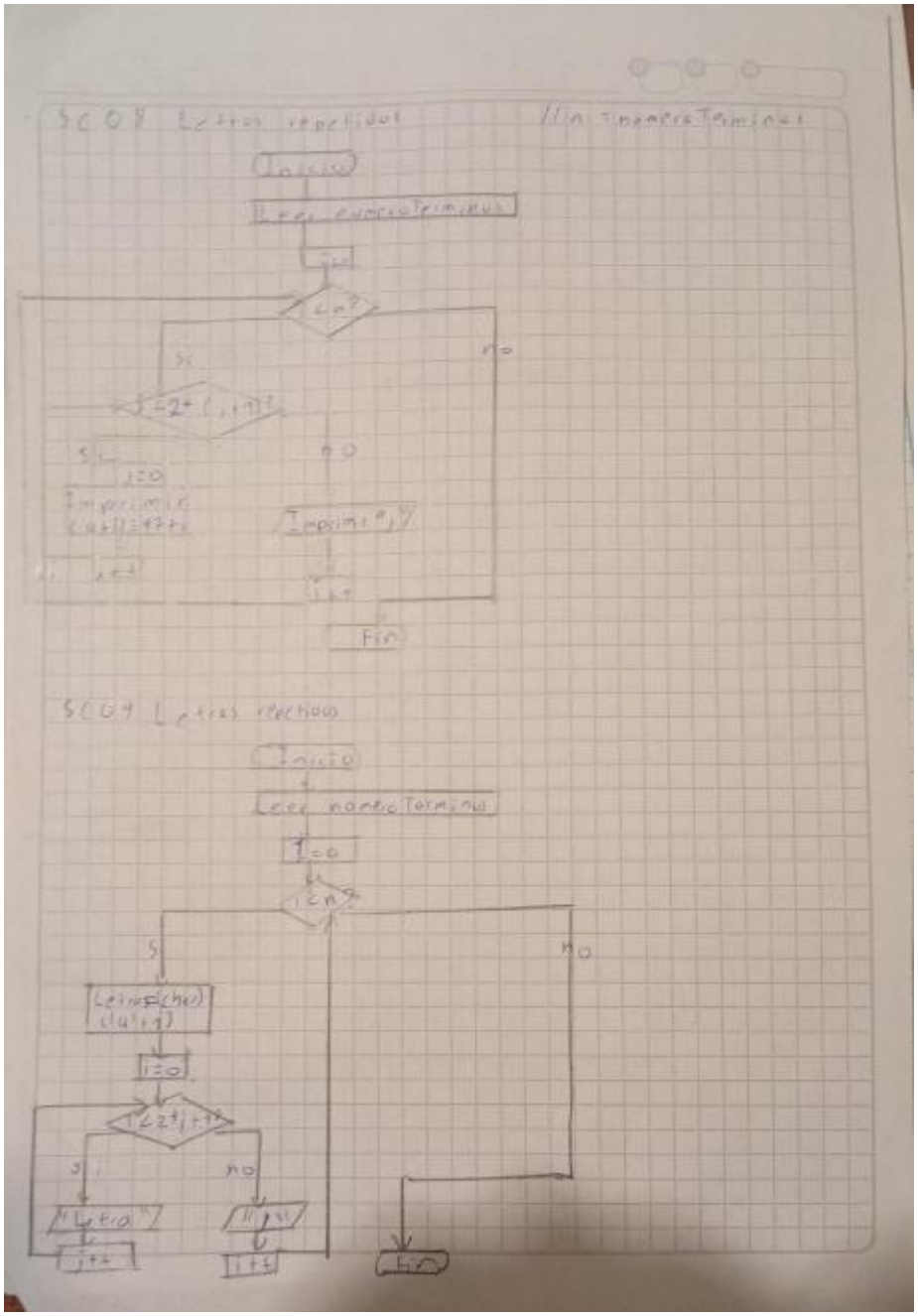

Ejercicio: Suma alternados
Inicio
Lee numeroTerminos
i = 0
i < n ?
si
no
% 2 == 0 ?
si
no
Fin
Ejercicio: SC02 : Etapa 1 seguro
Inicio
Lee numero Terminos
i = 0
i < n ?
si
no
Fin


Inicio
Leer numero Terminos
a=2, b=3, i=0
i<n
Si
no
i=0
Salida resultado
i<a-2
Si
no
a+b
i++
a=b
b=c
i++
Fin


Ejercicio 3 CO 4 : 0 pares 33 ciclicos // n = numero terminos
Inicio
Leer numero terminos
operadores = { '+', '-', '*', '/' };
i < n ?
si
no
Imprimir operadores [i % 4]
i = i + 1
Fin
Ejercicio 3 CO 5 Tarjetas diferentes ciclicos
Inicio
Leer numero terminos
diferentes = { "♠", "♣", "♥", "♦" };
i = 0
i = n ?
si
no
Imprimir diferentes [i % 3]
i = i + 1
Fin


1C06 Letra (Serie, 4 (vo)
// n : = numero Terminos
Inicio
Leer numero Terminos
i = 0
i < n
si
no
Imprimir (char)
(65+i) = 97 + i
i + 1
FIN
SLD7 Letra y signo alternados
Inicio
Leer numero Terminos
i = 0, signo = '+'
i < n
si
no
Imprimir (char)
(65+i) = 97 + i
Imprimir signo
signo = = '+'
si
no
signo = '-'
signo = '+'
i + 1
Imprimir salto de Linea
Fin


SC08 Letras repetidas
//n = numero Terminos
Inicio
Leer numeroTerminos
i=1
i<n?
SI
NO
Imprimir
Imprimir "y"
i=i+1
Fin
SC09 Letras repetidas
Inicio
Leer numeroTerminos
I=0
i<n?
SI
NO
i=0
Imprimir "Letra"
j++
i++
Fin

SC10 LETRAS CON REPETICION

a, bb, ccc, dddd, eeeee for

1 2 3 4 5

for a-z 1a 2bb 3ccc for

char caracter = (char)(a + i)

i = 0 -> 97+0 = 97 -> a
i = 1 -> 97+1 = 98 -> b
i = 2 -> 97+2 = 99 -> c
i = 27 -> 97+27 = 124 -> {

a b c d e f g
h i j k l m n
o p q r s t u
v w x y z

for i < NUMERO TERMINOS

-> i % 26

SC11: GRUPOS DE #

for IMPRIMIR j < i

+2 +2 +2

2 4 6 8 for AVANZAR i

SC12: VOCALES REPETIDAS

a, ee, iii, oooo, uuuuu, aaaaaa for

1 2 3 4 5 6

for a-z

char = {a, e, i, o, u}
-> VOCALES

(2) caracter = VOCALES % 5

{a, e, i, o, u}
1 2 3 4 5

SC13: LETRA REPETIDA CRECIENTE

A, BB, CCC, DDDD

1 2 3 4

MAYÚSCULA

IGUAL QUE SC10 SOLO CAMBIO char caracter = (char)(A + i)

Tema | Día | Mes | Año | ESTILO

SC14. ALFABETO

for -> A-Z

char ALFABETO = (char)(A + (i%26))

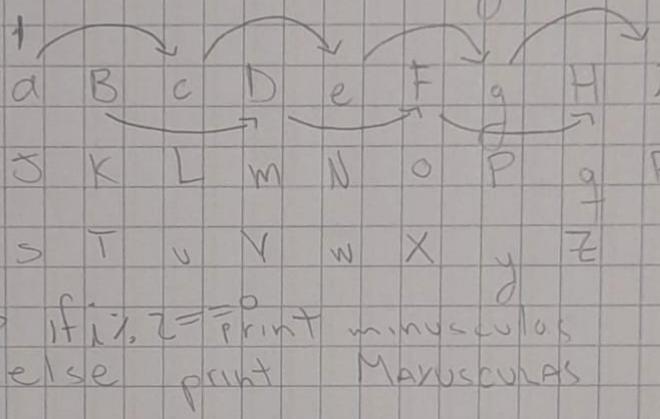
SC15 BLOQUES

*, **, ***, **** -> for AVANZAR i
1 2 3 4

└> FOR IMPRIMIR j<=i

SC16. ALTERNANCIA

: a, B, c, D, e, F



minúsculas impares

Mayúsculas pares

-> if i%2==0 print minusculos

└> else print MAYUSCULAS

SC17. SÍMBOLOS

+, -, *, /, %, +, -, * ...

for [+, %]

1 2 3 4 5

char SÍMBOLOS = {+, -, *, /, %}

caracter = SIMBOLOS %5

↪ ↑ BUCLE

PLANTEL: ASIGNATURA:

ALUMNO: Velez Jhostyn CURSO: 6R2CC

FECHA: CALIFICACIÓN: FIRMA PROFESOR:

TRIMESTRE:

SNO1. Fibonacci

ITERACIÓN	a	b	temp	Imprime
0	0	1	1	0
1	1	1	2	1
2	1	2	3	1
3	2	3	5	2

SNO2. IMPARES ALTERNADOS CON CERO.

Iteracion(i)	i%2=0	impar	Imprime
1	Falso	1	1
2	Verdadero	3	0
3	Falso	3	3
4	Verdadero	5	0

SNO3. FIBONACCI SOBRE IMPARES

Iteración(i)	a	b	impar	Imprime
0	0	1	1	0/1
1	1	1	3	1/3
2	1	2	5	1/5
3	2	3	7	2/7

SNO4. FIBONACCI SOBRE PARES

Iteración(i)	a	b	par	Imprime
0	0	1	2	0/2
1	1	1	4	1/4
2	1	2	6	1/6
3	2	3	8	2/8

SNO5. NUMEROS PRIMOS

Contador	Número (num)	¿ES Primo?	Imprime
0	2	Verdadero	2
1	3	Verdadero	3
2	4	Falso	(Nada)
2	5	Verdadero	5

SNO6 CUADRADOS PERFECTOS

Iteración (i)	Calculo	Imprime
1	1*1	1
2	2*2	4
3	3*3	9
4	4*4	16

SNO7. PROGRESIÓN ARIMETICA (+3)

Iteración (i)	val	val+3	Imprime
0	1	4	1
1	4	7	4
2	7	10	7
3	10	13	10

SNO8. PROGRESIÓN ARIMETICA (+5)

Iteración (i)	val	val+5	Imprime
0	3	8	3
1	8	13	8
2	13	18	13
3	18	23	18

SNO9 POTENCIAS

Iteración (i)	val	val*2	Imprime
0	2	4	2
1	4	8	4
2	8	16	8
3	16	32	16

Series numéricas

10. Potencias

3, 9, 27, 81, 243, 729, ...

$3 = 3^1$

$9 = 3^2$

$27 = 9 \cdot 3 = 3^2 \cdot 3 = 3^3$

$81 = 9 \cdot 9 = 3^2 \cdot 3^2 = 3^4$

$243 = 81 \cdot 3 = 9^2 \cdot 3 = 3^4 \cdot 3 = 3^5$

3	9	27	81	243	729
↓	↓	↓	↓	↓	↓
3^1	3^2	3^3	3^4	3^5	3^6

$$n = 3^{\sum_{i=1}^{m} i} \quad //$$

11. Numeros oblongos

2, 6, 12, 20, 30, 42, ... → a_n

4, 6, 8, 10, 12

2, 2, 2, 2

$a_n = an^2 + bn + c$

$2a = 2 \Rightarrow a = 1$ //

$3a + b = 4$

$3 + b = 4 \Rightarrow b = 1$ //

$a + b + c = 2$

$1 + 1 + c = 2 \Rightarrow c = 0$

$a_n = n^2 + n$

$a_n = n(n+1)$ //

12. Triangulares

1, 3, 6, 10, 15, 21, ... → a_n

2 3 4 5 6

1 1 1 1

$2a = 1 \Rightarrow a = \frac{1}{2}$

$a + b = 2$

$\frac{3}{2} + b = 2 \Rightarrow b = \frac{1}{2}$

$a + b + c = 1$

$\frac{1}{2} + \frac{1}{2} + c = 1 \Rightarrow c = 0$

$a_n = \frac{1}{2}n^2 + \frac{1}{2}n$

$a_n = \frac{1}{2}n(n+1)$ //

13. Factoriales

1, 2, 6, 24, 120, 720, ... → a_n

1! 2! 3! 4! 5! 6!

$1! = 1$

$2! = 2 \cdot 1 = 2$

$3! = 3 \cdot 2 \cdot 1 = 6$

$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$

+1

1 2 6 24 120 720 ...

n: 1 2 3 4 5 6

$\begin{cases} a_n = n \cdot a_{n-1}, & \text{si } n > 1 \\ a_1 = 1 \end{cases}$ //

$a_2 = 2 \cdot 1 = 2$ ✓ $a_3 = 3 \cdot 2 = 6$ ✓ $a_4 = 4 \cdot 6 = 24$ ✓

14. Signos alternos

1, -2, 3, -4, 5, -6, ...

n: 1 2 3 4 5 6

Si n es par es negativo

$$\begin{cases} a_n = n, & \text{si } n\%2 \neq 0 \\ a_n = -n, & \text{si } n\%2 == 0 \end{cases}$$

$1\%2 = 1 \neq 0 \Rightarrow a_1 = 1$

$2\%2 = 0 \Rightarrow a_2 = -2$

$3\%2 = 1 \Rightarrow a_3 = 3$

$4\%2 = 0 \Rightarrow a_4 = -4$

15. Cuadrados desplazados

n: 1 2 3 4 5 6

0, 2, 8, 18, 32, 50, ...

2 6 10 14 18

4 4 4 4

$2a = 4 \Rightarrow a = 2$

$3a + b = 2$

$6 + b = 2 \Rightarrow b = -4$

$a + b + c = 0$

$2 - 4 + c = 0 \Rightarrow c = 2$

$a_n = 2n^2 - 4n + 2$

$a_n = 2(n-1)^2$ // Porque n empieza en 1

16. Repunidades

1, 11, 111, 1111, 11111, ...

n: 1 2 3 4 5

$1 = \frac{9}{9} = \frac{10-1}{9}$

$11 = \frac{99}{9} = \frac{10^2-1}{9}$

$111 = \frac{999}{9} = \frac{10^3-1}{9}$

$1111 = \frac{9999}{9} = \frac{10^4-1}{9}$

$a_n = \frac{10^n-1}{9}$ //

17. Números de Lucas

n: 1 2 3 4 5 6

2, 1, 3, 4, 7, 11, ...

$a_1 = 2$

$a_2 = 1$

$a_n = a_{n-1} + a_{n-2}$

$a_3 = 1 + 2 = 3$

$a_4 = 3 + 1 = 4$

$a_5 = 4 + 3 = 7$

$a_6 = 4 + 7 = 11$

18. Crecientes

1, 2, 4, 7, 11, 16, ...

1 2 3 4 5

1 1 1 1

$2a = 1 \Rightarrow a = \frac{1}{2}$

$3a + b = 1$

$\frac{3}{2} + b = 1 \Rightarrow b = -\frac{1}{2}$

$a + b + c = 1$

$\frac{1}{2} - \frac{1}{2} + c = 1 \Rightarrow c = 1$

$a_n = \frac{1}{2}n^2 - \frac{1}{2}n + 1$

$a_n = \frac{1}{2}(n^2 - n + 2)$ //

$a_1 = \frac{1}{2}(1 - 1 + 2) = 1$

$a_2 = \frac{1}{2}(4 - 2 + 2) = 2$

$a_3 = \frac{1}{2}(9 - 3 + 2) = 4$

$a_4 = \frac{1}{2}(16 - 4 + 2) = 7$

19. Alternancias

1, 8, 9, 64, 25, 216, ...

n: 1 2 3 4 5 6

$1 = 1^2$

$8 = 2^3$

$9 = 3^2$

$64 = 4^3$

$25 = 5^2$

$216 = 6^3$

$$\begin{cases} a_n = n^2, & \text{si } n\%2 \neq 0 \\ a_n = n^3, & \text{si } n\%2 == 0 \end{cases}$$

ESTILO